
ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Department of Information Technology

Synopsis

On

TRAFFIX AI

Predictive Spillback-Aware Traffic Signal Control for Reducing Congestion, Fuel Consumption and CO₂ Emissions

1. INTRODUCTION

1.1 Problem Context (Real-world scenario)

In dense urban corridors such as the arterial roads of Indian cities, signalised intersections are often only a few hundred metres apart. When demand rises in peak hours, the queue at one intersection can grow longer than the road segment holding it. This is called queue spillback. The queue then blocks the upstream stop line, so vehicles that receive a green signal cannot discharge. Green time is wasted, the effective capacity of the junction falls, and the blockage spreads to neighbouring links and cross streets, producing cascading congestion and, in the worst case, gridlock.

Spillback also has an environmental cost. Vehicles caught in stop-and-go traffic and prolonged idling burn more fuel per kilometre and emit more CO₂. A signal controller that keeps downstream links from filling up can therefore reduce delay and may also reduce fuel use and emissions.

1.2 Motivation & Need

Fixed-time plans cannot react to rapidly changing or oversaturated conditions. Max-pressure control is popular for its decentralised design, but its effectiveness in oversaturated conditions is questionable and it needs sensors at every controlled intersection [4]. Recent controllers account for finite link storage [1, 2] and for spillback avoidance using predicted demand [3], and recent learning-based controllers account for emissions [6–8], but in the papers reviewed here the two concerns are treated separately. The emission-aware studies optimize a reward at a single intersection rather than preventing spillback, and their reported emission gains are modest and depend on traffic volume [7, 8]. There is a need for a controller that anticipates

spillback, acts before it happens, and is evaluated on spillback, fairness and emissions as well as delay.

2. LITERATURE SURVEY

The literature survey reviewed studies from 2021–2026 on traffic signal control, focusing on queue management, spillback prevention, prediction, and emissions. Existing methods address these aspects separately, with limited integration of queue prediction, downstream storage, and spillback prevention. The proposed work combines these features in a transparent rule-based controller and evaluates delay, spillback, fairness, fuel consumption, and CO₂ emissions. The identified gap is based on the reviewed studies and requires broader literature validation before claiming novelty.

TABLE 1. Existing Solutions and Limitations

S No.	Existing Solutions	What it Does	Limitations
1	Fixed-Time Control	Uses predefined signal timings	Cannot adapt to changing traffic.
2	Vehicle-Actuated Control	Adjusts signals using real-time traffic	Reactive; does not predict queues or storage.
3	Storage-Aware Control [1–4]	Considers downstream storage to reduce spillback	Focuses mainly on delay and queues
4	Prediction-Based Control [3]	Predicts short-term traffic demand.	Does not jointly consider queue growth and storage
5	Queue-Estimation [5]	Estimates current queue conditions	Limited proactive spillback prevention
6	Emission-Aware Control [6–8]	Consider fuel and CO ₂ emissions.	Does not explicitly address spillback

TABLE 2. Technologies Currently used

S No.	Technology	Purpose
1	SCOOT	Adaptive traffic signal coordination based on traffic conditions
2	SCATS	Real-time adaptive signal control for coordinated traffic networks
3	SUMO	Microscopic traffic simulation and controller testing
4	TraCI	Interface for connecting external signal controllers with SUMO
5	SUMO Emission Model	Estimates vehicle fuel consumption and CO ₂ emissions
6	Traffic Detectors	Provide vehicle, speed, occupancy, and queue information for signal control.

2.1 PROBLEM STATEMENT

Traffic congestion at closely spaced signalised intersections can lead to **queue spillback** when downstream road storage is insufficient for vehicles released from an upstream signal. This increases delay, stops, fuel consumption, and CO₂ emissions. Existing controllers may react to current traffic conditions without explicitly considering **short-term queue growth and downstream storage**.

This project develops a **rule-based controller** for a three-intersection corridor that predicts queue growth and compares the expected vehicle release with available downstream storage. When storage is insufficient, the controller limits the upstream green phase to reduce spillback while maintaining acceptable cross-street traffic flow. Its performance is compared with **fixed-time, actuated, and max-pressure control** under different demand conditions.

3. PROPOSED SOLUTION OVERVIEW

- **Real-Time Vehicle Detection**
 - Vehicles will be detected in real time using CCTV or video feeds.
 - The system will identify different vehicle types such as cars, motorcycles, buses, trucks, and auto-rickshaws.
- **Vehicle Classification**
 - The AI model will classify detected vehicles based on their type.
 - Vehicle classification will be used to estimate the overall traffic load more accurately.

- **Mixed Traffic Load Estimation**

Instead of considering only the total number of vehicles, the system will consider:

- Vehicle type
- Vehicle count
- Approximate road-space occupancy
- This will provide a more realistic estimation of traffic load in mixed-traffic conditions.

- **Queue Length Estimation**

- The system will continuously estimate the length of the vehicle queue on each approach.
- The queue growth rate will also be monitored to identify rapidly increasing congestion.

- **Spillback Risk Detection**

- The system will identify the risk of traffic spillback when downstream occupancy and upstream queue length reach critical levels.
- For example, 90% or higher downstream occupancy can be classified as a high spillback-risk condition.

- **Early Spillback Prediction**

- The system will analyze the rate at which the queue is growing.
- Based on queue growth and downstream capacity, it will predict whether the queue is likely to reach the upstream intersection.
- This allows preventive action to be taken before actual spillback occurs.

- **Adaptive Traffic Signal Control**

- Signal timings will be dynamically adjusted based on the detected spillback risk.
- When an approach has a high spillback risk, the system can temporarily reduce incoming traffic by modifying its signal timing.
- Other approaches with available downstream capacity can be given appropriate green time.

- **Spillback Prevention**

- The system will prevent unnecessary vehicles from entering a congested downstream road when its capacity is nearly exhausted.
- This will help reduce intersection blockage, queue overflow, and gridlock.

- **Emission Estimation**

- The controller does not optimise CO₂ directly. By preventing spillback and reducing stop-and-go driving and idling, it is expected to lower fuel consumption and CO₂ emissions indirectly.

4. OBJECTIVES

1. Build a synthetic SUMO corridor of three signalised intersections with 100–200 m links and medium and high demand scenarios, with teleporting disabled so that spillback is visible.
2. Implement the predictive spillback-aware controller (queue prediction, expected release versus free downstream storage, protective control with a maximum hold time).
3. Compare it with fixed-time, actuated and max-pressure control over 10 random seeds per scenario. The target is lower mean spillback duration than the best baseline at high demand, with 95th-percentile cross-street delay no more than 10% worse than that baseline.
4. Quantify network-wide fuel consumption and CO₂ emissions using the HBEFA trip-level output, report them as relative differences, and relate them to stops per vehicle and speed profile.
5. Test sensitivity to the risk threshold (for example 70%, 80% and 90% occupancy) and to noise in the queue estimate.

5. CONTRIBUTIONS

1. A predictive rather than reactive spillback-prevention rule that compares the expected release of each phase with the free downstream storage.
2. A cross-street safeguard (maximum hold time) evaluated using tail-delay and cross-street queue metrics.
3. A broader evaluation than average delay: spillback duration and frequency, blocked vehicle-hours, fairness indicators, fuel and CO₂.
4. A reproducible open-source SUMO/TraCI framework with sensitivity analysis. Results are simulation-based on a small synthetic network and are not claimed as field-validated.

6. THEORETICAL ANALYSIS

6.1 Block Diagram

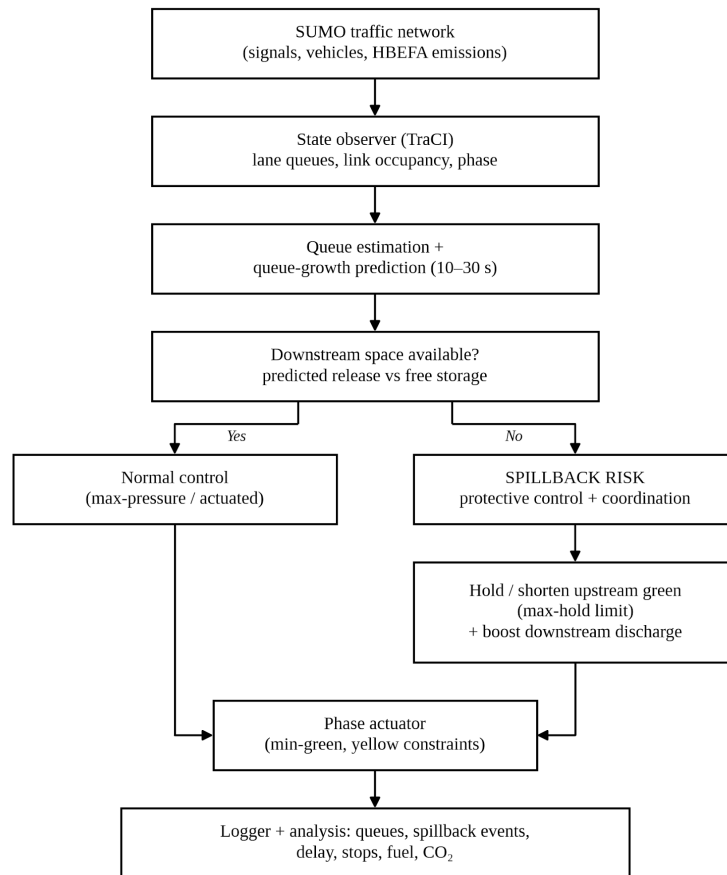


Fig. 1: Block diagram of the proposed spillback-aware controller

6.2 Hardware / Software Design

Table 3. Hardware requirements

Component	Specification	Purpose
Processor	Intel Core i5 (10th gen) / AMD Ryzen 5 or better, 4+ cores	Run SUMO and the Python controller; parallel runs of different seeds
Memory (RAM)	8 GB minimum, 16 GB recommended	Batch experiments (controller $\times$ demand $\times$ seed)
Storage	20 GB free disk space (SSD preferred)	Network files, raw outputs and logs of many runs
Display	1920 $\times$ 1080 monitor	Observe SUMO-GUI and plots

Table 4. Software requirements

Category	Software	Purpose
Operating system	Windows 10/11 or Ubuntu 20.04+	Platform for simulation and development
Traffic simulator	SUMO with netedit and SUMO-GUI	Microscopic traffic simulation and network design
Emission model	HBEFA model built into SUMO	Per-vehicle fuel and CO ₂
Control interface	TraCI and sumolib (Python)	Read network state and set signal phases
Programming language	Python 3.x	Observer, predictor, controller, experiment runner
Libraries	NumPy, pandas, Matplotlib	Computation, result tables, plots
Development tools	VS Code / Jupyter Notebook	Coding and analysis
Version control	Git and GitHub	Code management and repository link

7. APPLICATIONS

1. **Urban Arterials:** Prevents queue spillback between closely spaced intersections.
2. **Grid Networks:** Helps manage congestion across interconnected signalised junctions.
3. **High-Risk Locations:** Useful near **flyovers, markets, schools, and industrial areas**.
4. **Smart Cities:** Supports adaptive traffic management and smoother traffic flow.
5. **Emission Reduction:** Helps reduce stop-and-go traffic, fuel consumption, and CO₂ emissions.
6. **Traffic Planning:** Enables planners and researchers to test signal strategies in **SUMO** before field deployment.

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Project GitHub Repository Link: _____

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